

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Start TxFFE on Asymmetric Transitions
Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

Add the requirement to set the Start TxFFE bit in the Link Partner when doing the Asymmetric Transition before executing the TxFFE negotiation.

Benefits as a result of the changes:

A full description of the transition flow.
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An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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None

An analysis of the hardware implications:
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None

An analysis of the software implications:
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None

An analysis of the compliance testing implications:
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Need to add a check that the bit is set as part of Asymmetric Transitions.
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Actual Change

(a). 4.2.2.5.1 -

To Text:

7. The new transmitter (Tx2) shall start transmitting back-to-back Gen 4 TS1 with the *Indication* field set to 2h. The new transmitter shall set the *Start TxFFE* bit in the Tx Status byte of the Gen 4 TxFFE register in the SB Register Space of the adjacent Router/Re-timer and execute the Gen 4 Transmitter flow for TxFFE negotiation as described in Section 4.1.2.5.1.1.2. Since all transmitters are using the same clock source, the new transmitter (Tx2) transmits the Gen 4 Training Sequences with the common SSC modulation.

(b). 4.2.2.6.2 -

To Text:

7. The new transmitter (Tx1) shall start transmitting back-to-back Gen 4 TS1 with the *Indication* field set to 2h. The new transmitter shall set the *Start TxFFE* bit in the Tx Status byte of the Gen 4 TxFFE register in the SB Register Space of the adjacent Router/Re-timer and execute the Gen 4 Transmitter flow for TxFFE negotiation as described in Section 4.1.2.5.1.1.2. Since all transmitters are using the same clock source, the new transmitter (Tx1) transmits the Gen 4 Training Sequences with the common SSC modulation.